Stock market analysis in SQL

Insights report: moving-average signals on five NSE stocks, January 2015 to July 2018

Prepared by Aman Tiwari | Stocks: Eicher Motors, Hero Motocorp, Infosys, TCS, TVS Motors | 889 trading days each

Summary

- **Two of the raw numbers are wrong.** TCS (31 May 2018) and Infosys (15 June 2015) each show a one-day fall of about 50%. Both are 1:1 bonus issues, not crashes. On raw prices TCS looks like -23.8% and Infosys like -30.9%; adjusted, they are **+52.4%** and **+38.2%**.
- **The winners are TVS (+86.9%) and Eicher (+82.6%)**, followed by TCS, Infosys and, far behind, Hero (+6.0%). The two best performers are also the most volatile and had the deepest drawdowns (TVS -34.6%, Eicher -28.3%).
- **The golden-cross rule did not beat holding.** Following the Buy/Sell signals lost money on Hero (-16.3%) and TCS (-22.4%), and earned less than buy-and-hold on four of five stocks. Only Eicher did better with the signals (+76.5% vs +52.2%).
- **Signals whipsaw.** 36 of 91 signals came within 30 trading days of the previous opposite signal, and following the first of each pair lost 5 to 7% on average.
- **Latest signals disagree with the long-run picture for TVS and Eicher** (Sell after big gains), and agree for Infosys and TCS (Buy, strong 12-month trend). The raw-data TCS "Sell" of 5 June 2018 is an artefact of the bonus issue.

Scorecard (adjusted prices)

Stock	Period return	12-month return	Volatility (ann.)	Max drawdown	Signals (Buy / Sell)	Latest signal
TVS Motors	+86.9%	-9.8%	30.8%	-34.6%	8 / 8	Sell (17 May 2018)
Eicher Motors	+82.6%	-4.3%	29.1%	-28.3%	6 / 7	Sell (6 Jun 2018)
TCS	+52.4%	+56.3%	21.6%	-24.1%	12 / 12	Buy (20 Apr 2018)
Infosys	+38.2%	+40.4%	23.8%	-31.1%	10 / 10	Buy (7 May 2018)
Hero Motocorp	+6.0%	-11.2%	22.3%	-26.9%	9 / 9	Sell (22 May 2018)

12-month = last 252 trading days. Volatility = standard deviation of daily returns x sqrt(252). Signals use unrounded 20-day and 50-day averages on adjusted prices. Past performance in this sample says nothing certain about the future; this is a coursework analysis, not investment advice.

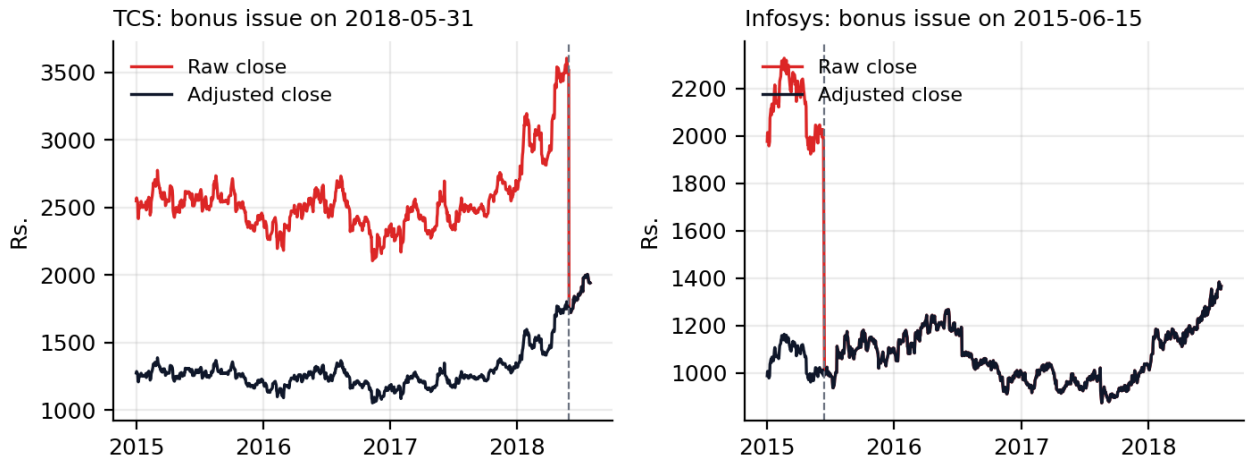
1. Data and method

- **Data.** Daily NSE prices for five stocks, 1 Jan 2015 to 31 Jul 2018, 889 rows each, same columns in every table. Only **close_price** is used.
- **Moving averages.** Simple 20-day and 50-day averages built with window functions (frame of 19 and 49 preceding rows plus the current row). The first 19 / 49 rows are left NULL because a partial window is not a true average.
- **Signal rule.** Buy on the day the 20-day average crosses above the 50-day average; Sell on the day it crosses below; Hold otherwise. A cross compares today with yesterday (LAG), so only the single day that changes counts.
- **Case study.** TVS Motors is the single-stock walk-through (tables tvs1, tvs2). Task 10 repeats the logic for all five stocks in one query using PARTITION BY.
- **Data quality.** The only missing values are deliverable_qty on 9 Dec 2015 (four stocks) and 31 Aug 2017 (Infosys). Because the gaps hit several unrelated companies on the same date, the exchange report is the likelier source, not the companies. Close prices are complete, so no analysis here is affected.

Where each number comes from

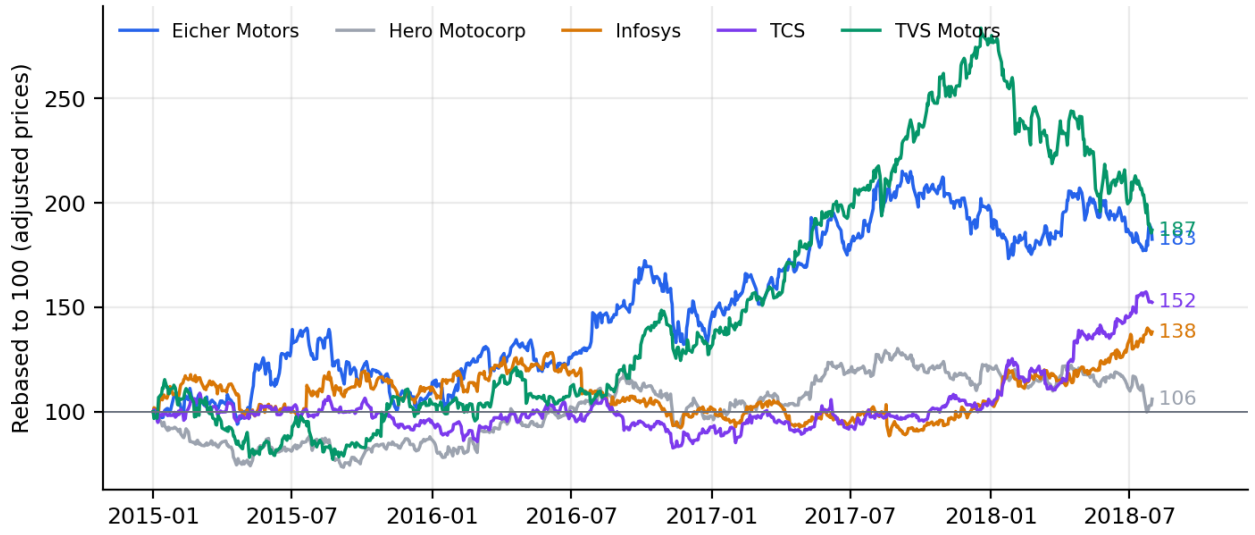
Section of report	SQL file query
Period returns (raw)	Task 11
Worst day / data trap	Task 12
Adjusted returns	Task 13 and B1
Signal counts, latest signal	Task 10 (raw prices); adjusted version in B2/B4 logic
Fake / hidden signals	B2 (TCS), Python cross-check for Infosys
Whipsaws	B3
Follow-the-signals result	B4
Volatility, drawdown, 12-month return	Python cross-check (pandas) on the same adjusted prices

2. The data trap: two price events



1. TCS and Infosys "crashed" because of bonus issues, not losses	
Claim	Both stocks show a one-day fall of about 50% that is not a loss of value. In a 1:1 bonus issue the share count doubles and the price halves overnight. Prices before the event must be divided by 2 to be comparable.
Evidence	Task 12: TCS worst day 31 May 2018 at -50.4% (close 1,744.80); Infosys worst day 15 Jun 2015 at -49.9% (close 991.10). Every other stock's worst day is between -6.2% and -9.7%. Task 13 adjusts both.
Caveat	I identified the events from the size of the jump and the standard meaning of a 1:1 bonus; I did not verify them against company announcements. The factor 2 assumes a 1:1 ratio exactly. Other corporate actions (dividends, any smaller splits) are not adjusted.

3. Performance and risk



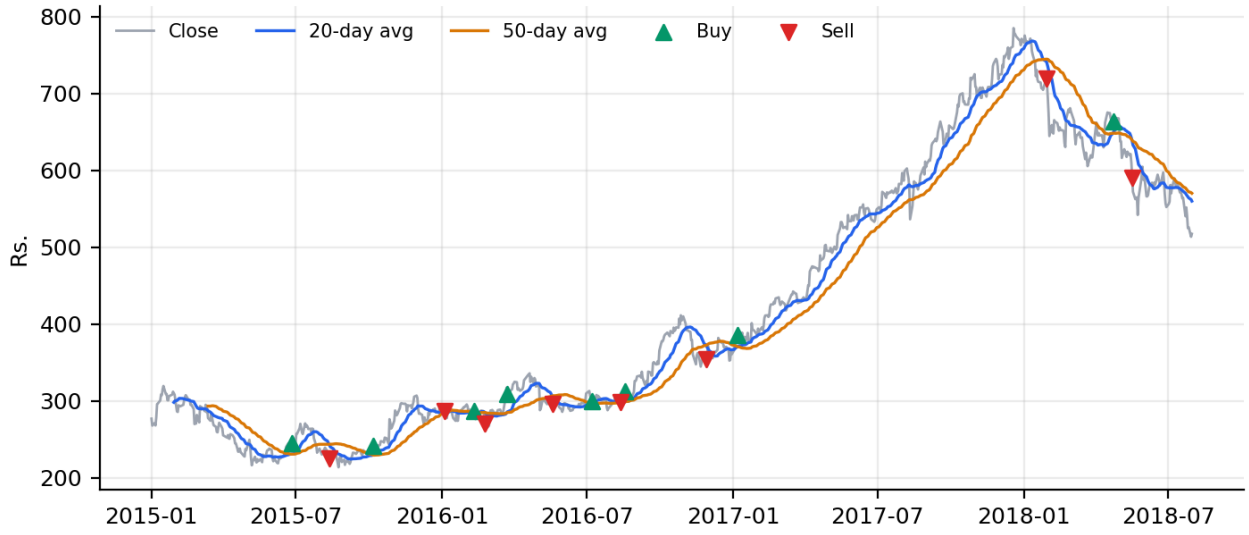
Stock	First close	Last close	Raw change	Adjusted change	Volatility	Max drawdown (date)
TVS Motors	276.85	517.45	+86.9%	+86.9%	30.8%	-34.6% (2018-07)
Eicher Motors	15,239.15	27,820.95	+82.6%	+82.6%	29.1%	-28.3% (2015-11)
TCS	1,274.10	1,941.25	-23.8%	+52.4%	21.6%	-24.1% (2016-11)
Infosys	987.90	1,365.00	-30.9%	+38.2%	23.8%	-31.1% (2017-08)
Hero Motocorp	3,107.30	3,293.80	+6.0%	+6.0%	22.3%	-26.9% (2015-09)

First close for TCS and Infosys is the adjusted (halved) value.

2. The winners and losers list changes after adjustment	
Claim	On raw prices TCS and Infosys appear to be the two losers. After adjustment they are clearly positive; the only weak stock left is Hero. The ranking becomes TVS, Eicher, TCS, Infosys, Hero.
Evidence	Task 11 (raw): TCS -23.8%, Infosys -30.9%. Task 13 / B1 (adjusted): TCS +52.4%, Infosys +38.2%. TVS (+86.9%), Eicher (+82.6%) and Hero (+6.0%) are unaffected because they had no event.
Caveat	Start-to-end percentage change depends on the two endpoint days. 2018 has only seven months of data, so it should not be called a "best year" in any yearly average (TCS Task 3: 2016 average 2,419.00). Returns exclude dividends.

3. Higher return came with higher risk	
Claim	The two best performers were also the most volatile and had the deepest falls from a peak. TVS fell 34.6% from its peak to its low on 30 Jul 2018, even though it is up 86.9% over the full period.
Evidence	Volatility: TVS 30.8%, Eicher 29.1%, Infosys 23.8%, Hero 22.3%, TCS 21.6%. Max drawdown: TVS -34.6%, Infosys -31.1%, Eicher -28.3%, Hero -26.9%, TCS -24.1%.
Caveat	Annualised volatility uses 252 trading days. With under four years of data, drawdown figures are dominated by one or two episodes (the 2015 and 2018 sell-offs). Not a long-run risk estimate.

4. Signals per stock



TVS Motors: close, 20-day and 50-day averages, with Buy and Sell days marked.

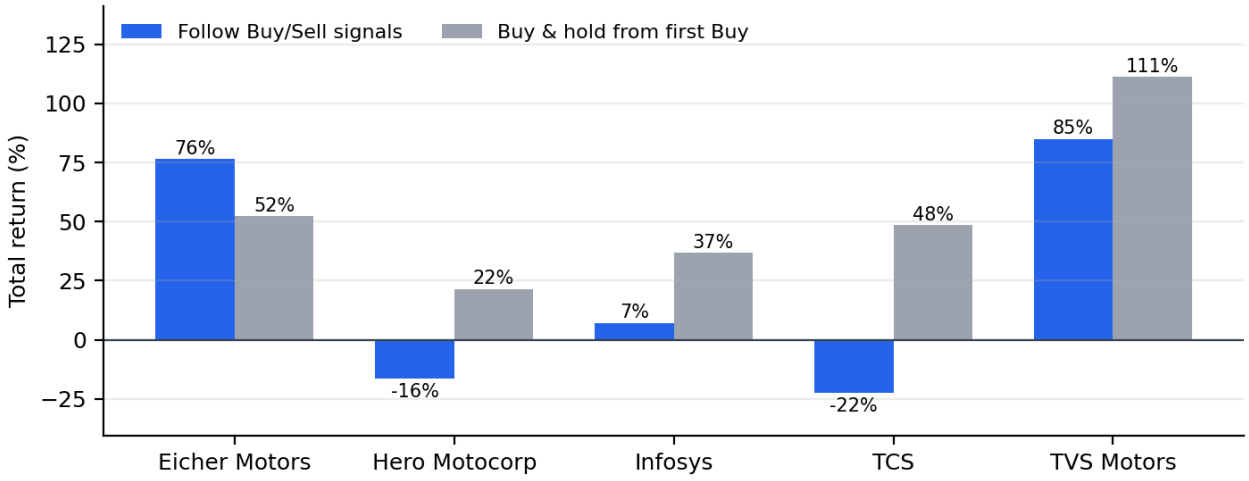
Stock	Buys	Sells	Latest signal	Raw-price signals that are fake	Signals hidden by the price cliff
TVS Motors	8	8	Sell, 17 May 2018	none	none
Eicher Motors	6	7	Sell, 6 Jun 2018	none	none
TCS	12	12	Buy, 20 Apr 2018	2018-06-05 Sell	none
Infosys	10	10	Buy, 7 May 2018	2015-08-18 Buy	2015-07-01 Buy, 2015-07-08 Sell, 2015-07-27 Buy
Hero Motocorp	9	9	Sell, 22 May 2018	none	none

Counts are on adjusted prices. On raw prices the Buy/Sell counts are Eicher 6/7, Hero 9/9, Infosys 9/9, TCS 12/13, TVS 8/8 (Task 10).

4. The bonus-issue cliff creates false signals and hides real ones	
Claim	Raw prices distort the moving averages for up to 50 trading days after a cliff. On TCS this produced a Sell on 5 Jun 2018 that does not exist on adjusted prices; on Infosys it produced a Buy on 18 Aug 2015 and hid three real signals in July 2015.
Evidence	B2 compares TCS signals on raw and adjusted prices: 24 of 25 are identical; the single difference is the 5 Jun 2018 Sell (raw only). The Infosys comparison was done with the same logic in a Python cross-check: raw-only Buy 18 Aug 2015; adjusted-only Buy 1 Jul, Sell 8 Jul, Buy 27 Jul 2015.
Caveat	Only TCS and Infosys were adjusted. If other stocks had unflagged corporate actions smaller than a 50% jump, the worst-day test would not catch them. The Infosys check is not in the SQL file; B2 can be copied for Infosys by changing the table and event date.

5. The signal count is small, so conclusions rest on few events	
Claim	Each stock produced only 13 to 24 signals in 3.5 years, so any average built from them is fragile.
Evidence	Adjusted Buy/Sell counts: Eicher 6/7, Hero 9/9, Infosys 10/10, TCS 12/12, TVS 8/8 (91 signals in total).
Caveat	Eicher and TVS signal results rest on 6 to 8 trades each. One large trade (TVS +86.7%) drives its total.

5. Would following the signals have paid?



Long-only strategy: buy at each Buy close, sell at the next Sell close, open trade marked at 31 Jul 2018. Buy-and-hold starts on the first Buy day for a fair comparison.

Stock	Trades	Won	Avg trade	Worst	Best	Quick reversals (30 days)	Avg result of the first signal in those pairs
TVS Motors	8	3	+11.2%	-11.0%	+86.7%	7	-6.43%
Eicher Motors	6	6	+10.2%	+3.1%	+25.1%	1	-5.03%
TCS	12	3	-1.9%	-11.0%	+14.0%	15	-6.87%
Infosys	10	4	+1.2%	-10.4%	+21.4%	9	-6.05%
Hero Motocorp	9	3	-1.7%	-11.4%	+8.1%	4	-7.19%

Source: SQL B3 (quick reversals) and B4 (trade results), adjusted prices.

6. Following the signals cost money on average	
Claim	Across the four stocks other than Eicher the rule underperformed holding; on Hero and TCS it lost money outright. Eicher is the exception, where every one of its 6 trades was a winner.
Evidence	B4 compounded results: Eicher +76.5% (buy-and-hold from first Buy +52.2%), TVS +85.0% (+111.1%), Infosys +7.1% (+36.8%), Hero -16.3% (+21.6%), TCS -22.4% (+48.4%). Hero and TCS won only 3 of 9 and 3 of 12 trades.
Caveat	No brokerage, taxes or slippage are included; they would make every strategy number worse. Compounded returns assume all capital is reinvested in each trade. The comparison depends on one 3.5-year window, which included a strong uptrend (good for buy-and-hold) in most of these stocks.

7. Quick flip-flops are where the damage happens	
Claim	Many signals reverse within about six weeks. A trader following the first signal of each such pair lost 5 to 7% on average per pair.
Evidence	B3: 36 quick reversals across five stocks (TCS 15, Infosys 9, TVS 7, Hero 4, Eicher 1). Average result of following the first signal: Eicher -5.03%, Hero -7.19%, Infosys -6.05%, TCS -6.87%, TVS -6.43%. TCS (adjusted) had the longest flip-flop stretch: 15 of its 24 signals came within 30 trading days of the previous one.
Caveat	Thirty trading days is my own threshold; a different cut-off changes the counts. Eicher's average comes from one pair only.

6. Does the latest signal agree with the long-run trend?

Stock	Latest signal	State on 31 Jul 2018	Close vs 50-day avg	12-month return	Period return	Agrees with long-run trend?
TVS Motors	Sell (17 May 2018)	20d<50d	-9.3%	-9.8%	+86.9%	No: Sell after an +86.9% run; recent trend is down
Eicher Motors	Sell (6 Jun 2018)	20d<50d	-3.4%	-4.3%	+82.6%	No: Sell after +82.6%; recent drift is mildly negative
TCS	Buy (20 Apr 2018)	20d>50d	+4.4%	+56.3%	+52.4%	Yes: Buy; 12-month +56.3%
Infosys	Buy (7 May 2018)	20d>50d	+6.2%	+40.4%	+38.2%	Yes: Buy; 12-month +40.4%
Hero Motocorp	Sell (22 May 2018)	20d<50d	-6.2%	-11.2%	+6.0%	Yes: weak long-run and weak recently

8. For TVS and Eicher the Sell signal and the long-run trend point in opposite directions	
Claim	Both stocks gained more than 80% over the period but are now trading below their 50-day average with a negative 12-month return, and both are in a Sell state. I would weigh the signal as a short-term caution about momentum, not as evidence that the long-run gain has reversed.
Evidence	TVS: close 9.3% below its 50-day average, 12-month -9.8%, Sell on 17 May 2018. Eicher: close 3.4% below, 12-month -4.3%, Sell on 6 Jun 2018. Both still +82 to +87% from the start (Task 13 / B1).
Caveat	Whether to trust the signal or the trend depends on the investor's horizon. The signal lags (see section 7), so by the time it fires part of the fall has already happened.

9. For TCS the correct latest signal is a Buy, not the Sell shown on raw data	
Claim	On adjusted prices TCS's last real signal is the Buy on 20 Apr 2018, which agrees with its +56.3% 12-month return and with the 20-day average being above the 50-day. The raw-data Sell of 5 Jun 2018 should be ignored.
Evidence	B2 shows the 5 Jun 2018 Sell exists only on raw prices. On 31 Jul 2018 TCS closes 4.4% above its 50-day average (adjusted).
Caveat	If an analyst used the raw Task 10 output without checking the price cliff, the last-signal column would be wrong for TCS. Infosys has the same exposure earlier in the sample (2015).

7. Questions about the method

A moving average only uses past prices. How early can a golden cross tell you anything?

Not early. A 50-day average has a built-in lag of roughly 25 days, and the 20-day one about 10, so a crossover can only happen after the price has already moved for several weeks. That lag means Sells often fire after a drop has happened and Buys after a rise has happened, which shows up in the trade results in section 5. The 50-day average only exists from the 50th trading day (13 March 2015), so the first ten weeks of the sample cannot produce any signal.

Which conclusions change after the price correction?

- TCS and Infosys go from "the two losers" (-23.8%, -30.9%) to winners (+52.4%, +38.2%).
- TCS: one raw Sell (5 Jun 2018) disappears, so its last signal changes from Sell to Buy (20 Apr 2018). TCS counts move from 12/13 to 12/12.
- Infosys: counts move from 9/9 to 10/10; one raw Buy (18 Aug 2015) disappears and three July 2015 signals appear.
- Eicher, Hero and TVS are unchanged. I did not have the course deck, so the specific deck figures still need to be compared against these numbers before you submit.

What is not in this data that a real investor would need?

- **Dividends** (total return is higher than price return) and other corporate actions beyond the two I adjusted.
- **Costs:** brokerage, taxes and the bid-ask spread. With 8 to 12 trades per stock these would be material.
- **News and fundamentals:** earnings, regulation, the 2016 demonetisation and the 2017 GST roll-out all fall inside this window.
- **The broader market:** there is no index in the data, so I cannot separate stock-specific moves from the whole market moving. No risk-free rate either.

8. Conclusion

The most important finding is about the data, not the signals: two of five series contained a price event that flipped their story from loss to gain and polluted their signals. After correcting it, TVS and Eicher led the group but with large drawdowns, and the golden-cross rule did not add value over holding on four of five stocks, mainly because of frequent reversals. The method is simple and transparent, which makes it a good teaching tool, but on this evidence it is a weak standalone trading rule.